

ALTOONA-BLAIR COUNTY AP, PA, USA

WMO: 725126

Lat: 40.296N

Lon: 78.320W

Elev: 1480

StdP: 13.93

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.5	9.8	-5.5	4.3	7.9	-1.2	5.4	11.8	28.1	30.1	25.9	28.5	9.9	300	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	88.2	71.9	85.6	70.8	83.0	69.7	74.9	84.1	73.3	81.8	71.9	79.6	8.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	125.5	79.6	70.5	118.6	77.8	69.4	114.2	76.5	39.2	84.0	37.8	81.8	36.5	79.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.9	20.1	17.8	DB	-0.8	92.7	6.6	3.8	-5.5	95.4	-9.4	97.6	-13.1	99.8	-17.9	102.5	(4)
			WB	-1.8	77.9	6.4	1.9	-6.4	79.2	-10.1	80.3	-13.7	81.3	-18.4	82.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	27.7	30.3	38.4	50.4	60.1	68.2	71.7	70.1	63.6	52.9	42.1	32.9
	DBStd	17.67	11.50	10.42	10.55	9.30	8.28	6.08	5.17	4.90	6.75	8.44	9.08	9.44
	HDD50	2629	696	556	388	105	11	0	0	0	1	66	268	538
	HDD65	5832	1156	972	826	444	199	39	7	13	106	386	688	996
	CDD50	2921	5	4	28	118	324	545	673	622	409	156	30	7
	CDD65	651	0	0	1	7	48	134	215	170	64	12	0	0
	CDH74	5011	0	0	12	84	414	987	1709	1237	497	70	1	0
	CDH80	1332	0	0	2	11	92	249	522	322	125	9	0	0
Wind	WSAvg	7.2	9.2	9.1	9.0	8.7	7.0	6.2	5.1	4.5	5.0	6.7	7.8	8.4
Precipitation	PrecAvg	36.4	2.2	2.2	3.2	3.4	3.7	3.7	3.5	3.0	3.3	2.6	3.3	2.4
	PrecMax	57.3	6.9	4.6	6.3	7.5	6.3	13.0	8.7	7.1	7.9	8.4	10.9	5.0
	PrecMin	24.3	0.3	0.6	0.5	0.5	0.4	0.6	1.0	0.0	0.3	0.1	0.6	0.4
	PrecStd	7.8	1.3	1.2	1.3	1.9	1.5	2.3	1.9	1.6	1.7	1.6	2.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	63.5	74.1	81.7	86.9	89.4	92.6	90.3	88.3	80.9	70.3	63.8
		MCWB	56.5	53.1	58.8	63.1	68.8	73.1	73.7	73.6	70.1	67.5	58.2	56.4
	2%	DB	56.0	55.7	67.0	76.1	82.4	85.8	88.6	86.5	83.8	75.2	64.2	57.1
		MCWB	51.3	48.0	54.4	59.5	66.4	71.1	72.5	71.7	68.9	64.0	54.7	51.7
	5%	DB	49.8	49.8	60.7	70.9	78.9	82.9	85.5	83.5	79.7	71.2	60.4	51.4
		MCWB	45.4	43.1	51.6	57.3	64.8	70.0	71.3	70.5	66.8	61.9	52.9	46.5
	10%	DB	43.3	44.8	55.1	66.1	74.7	79.9	82.5	80.9	76.0	67.0	56.5	45.9
		MCWB	39.5	39.5	47.4	54.4	63.2	68.4	70.2	69.3	65.7	59.5	50.3	41.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	55.6	61.3	66.2	71.4	76.0	77.0	76.4	74.1	70.1	61.0	58.0
		MCDB	60.7	61.4	72.1	76.3	82.4	84.4	88.1	85.8	82.7	76.8	66.0	62.8
	2%	WB	52.3	48.9	56.7	62.3	68.8	73.4	75.0	74.4	71.5	67.1	57.4	52.5
		MCDB	55.4	54.8	64.1	71.4	78.7	82.1	84.8	82.9	78.5	73.5	62.2	56.2
	5%	WB	45.6	43.9	52.7	59.4	67.0	71.8	73.4	72.6	69.7	63.6	54.4	46.9
		MCDB	49.2	48.8	59.3	68.1	76.1	79.9	82.2	80.2	75.9	68.2	58.8	50.2
	10%	WB	39.6	39.9	47.7	56.6	64.8	70.2	71.9	71.0	67.8	60.2	50.7	41.6
		MCDB	43.3	44.4	54.7	64.7	72.5	77.7	79.7	77.8	73.4	65.9	56.0	45.8
Mean Daily Temperature Range	5% DB	MDBR	13.1	14.5	16.7	19.9	19.3	18.4	19.4	19.5	19.4	17.8	15.0	12.4
		MCDBR	18.3	21.7	24.7	26.1	24.4	21.8	22.7	22.8	24.7	23.9	21.5	18.6
		MCWBR	15.2	16.2	15.7	14.6	11.7	10.2	9.6	10.4	11.5	13.3	14.7	14.8
	5% WB	MCDBR	17.5	19.6	22.5	22.1	21.0	19.0	18.7	19.1	18.6	19.2	18.4	17.3
		MCWBR	15.6	15.7	15.9	14.6	11.0	10.0	9.1	9.8	9.9	12.1	14.9	15.4
Clear-Sky Solar Irradiance	taub	0.307	0.322	0.347	0.373	0.410	0.463	0.480	0.474	0.392	0.364	0.320	0.303	
	taud	2.425	2.353	2.376	2.340	2.250	2.128	2.096	2.097	2.311	2.391	2.495	2.473	
	Ebn at Noon	271	283	287	286	276	260	254	252	268	264	265	264	
	Edh at Noon	26	32	35	38	43	49	50	49	37	30	24	23	
All-Sky Solar Radiation	RadAvg	530	802	1093	1452	1632	1801	1807	1614	1317	915	632	440	
	RadStd	43	100	89	134	163	150	116	92	163	103	69	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+104

Nomenclature: See separate page